LAB 3: Study and Simulation of Basic Networking Devices Using Cisco Packet Tracer

OBJECTIVE:

- To study the working of **hub, switch, bridge, router, and repeater**.
- To understand the **OSI layers** of different networking devices.
- To design and simulate a network using **Cisco Packet Tracer**.
- To verify communication between computers using **ping**.

THOERY:

1. Hub

A hub is a basic networking device used to connect multiple computers in a network. It operates at the **Physical Layer (Layer 1)** of the OSI model. When a hub receives data from any connected device, it broadcasts the data to all other devices. Due to this broadcasting nature, bandwidth is shared among all devices, which may lead to collisions and reduced network efficiency. In this experiment, the hub demonstrates how data is transmitted without filtering.

2. Switch

A switch is an intelligent networking device used to connect devices within a Local Area Network (LAN). It operates at the **Data Link Layer (Layer 2)** of the OSI model. The switch forwards data only to the intended destination by using MAC addresses, which improves network performance and reduces collisions. In the experiment, the switch allows efficient communication between multiple PCs.

3. Bridge

A bridge is used to connect two or more LAN segments and works at the **Data Link Layer (Layer 2)**. It filters network traffic by learning MAC addresses and allows only required data to pass between segments. This reduces unnecessary traffic and improves network performance. In this experiment, the bridge connects two switches and controls data flow between them.

4. Router

A **router** is a **network layer (Layer 3) device** used to connect **two or more different IP networks** and enable communication between them. Unlike switches, which work within the same network, a router forwards data packets between networks based on **IP addresses**.

A router has **multiple interfaces**, and **each interface is assigned a unique IP address** belonging to the network it connects to. These interface IP addresses act as the **default gateway** for the devices in their respective networks. When a device wants to communicate

with another network, it sends the data to its default gateway, and the router forwards the packet to the correct destination network.

In this experiment, the router connects two different networks: **192.168.1.0/24** and **10.0.0.0/24**. The router is configured with **192.168.1.1** on one interface and **10.0.0.1** on the other. Thus, the router enables successful communication between the two networks by routing packets through its interfaces.

5. Repeater

A repeater is a device used to extend the range of a network. It operates at the **Physical Layer (Layer 1)** of the OSI model. The repeater regenerates and retransmits weak signals to maintain signal strength over long distances. In this experiment, the repeater ensures reliable signal transmission between network segments.

CONFIGURATION:

1. Hub

Device Name	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0
PC2	FastEthernet0	192.168.1.3	255.255.255.0
PC3	FastEthernet0	192.168.1.4	255.255.255.0

2. Switch

Device Name	Interface	IP Address	Subnet Mask
PC0	FastEthernet0	192.168.1.1	255.255.255.0
PC1	FastEthernet0	192.168.1.2	255.255.255.0
PC2	FastEthernet0	192.168.1.3	255.255.255.0
PC3	FastEthernet0	192.168.1.4	255.255.255.0

3. Bridge

Device Name	IP Address	Subnet Mask
Switch0	192.168.1.2	255.255.255.0
Switch1	10.10.10.2	255.255.255.0
Bridge0	192.168.1.1	255.255.255.0

4. Router

Device Name	Interface	IP Address	Subnet Mask	Gateway
Router0	Gig0/0 (LAN 1)	192.168.1.1	255.255.255.0	N/A
Router0	Gig0/1 (LAN 2)	10.10.10.1	255.255.255.0	N/A
PC-LAN1	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC-LAN2	FastEthernet0	10.10.10.2	255.255.255.0	10.10.10.1

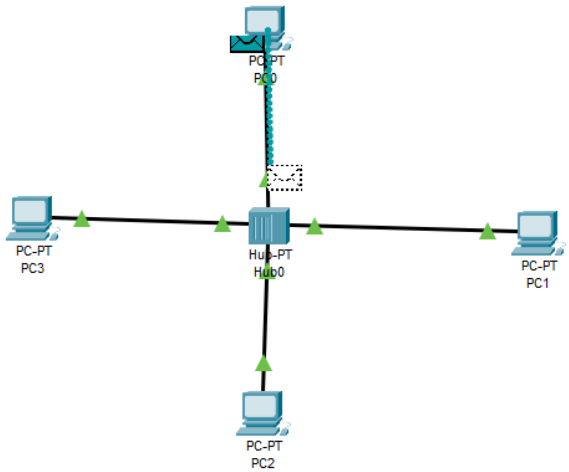
5. Repeater

Device Name	Connection Type	Status
Repeater0	Port 0 to Left Segment	Active
Repeater0	Port 1 to Right Segment	Active

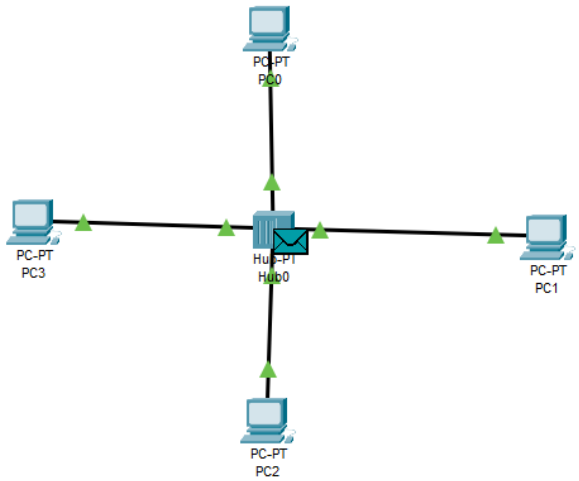
OUTPUT:

1. Hub

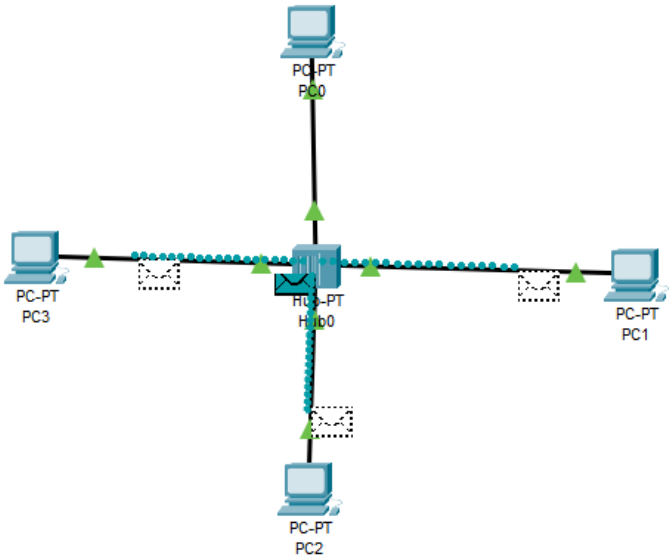
Step 1: PC0 sends a message



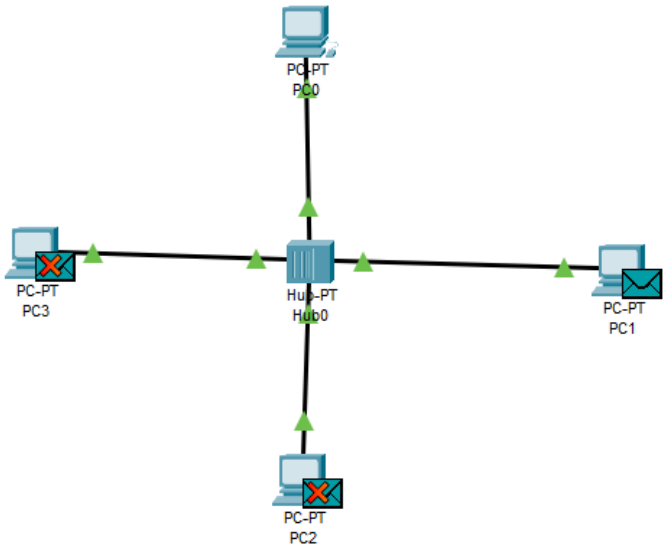
Step 2: Hub receives the message



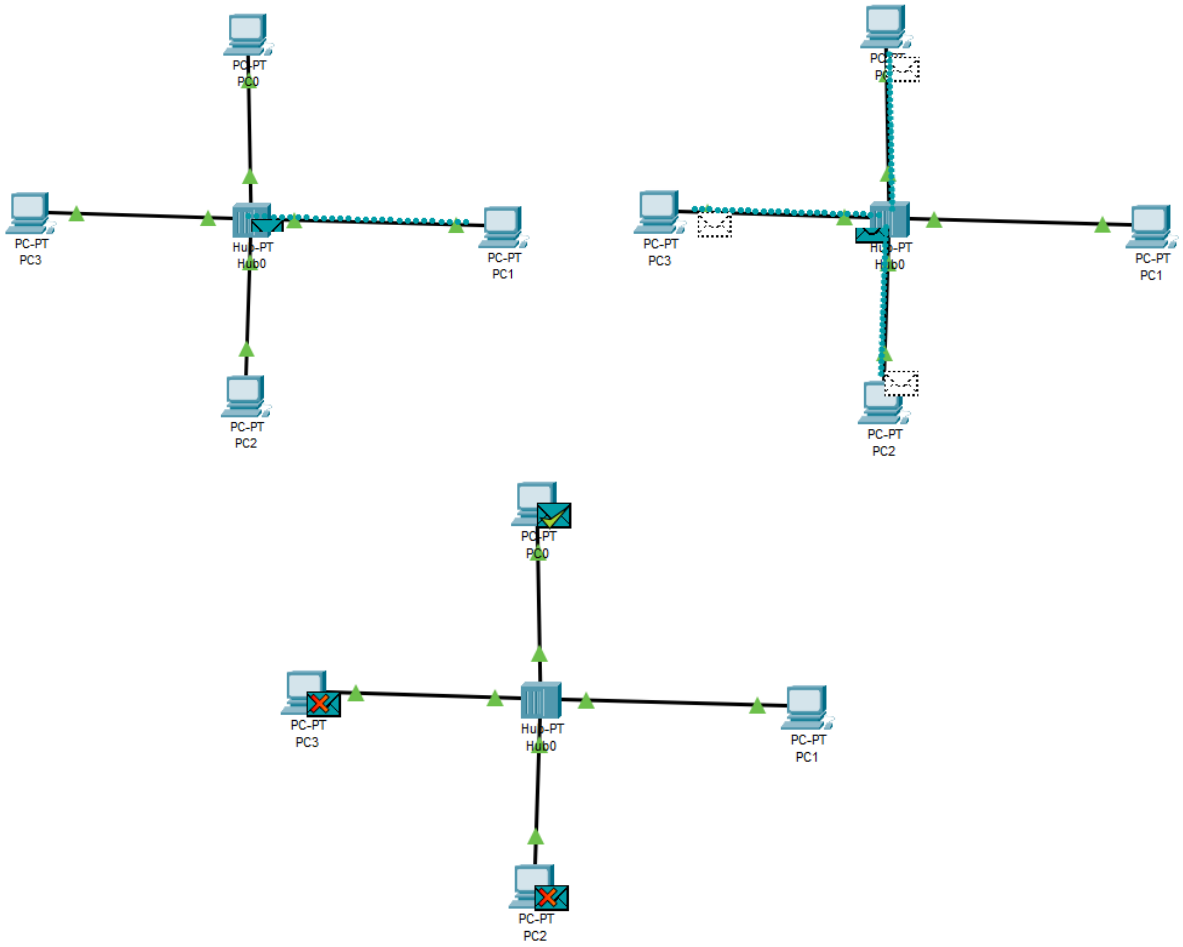
Step 3: Hub broadcasts the message



Step 4: PC1 receives the message, but other devices ignore the reply since it is not intended for them. (Shown in Packet Tracer by red X marks on their links)

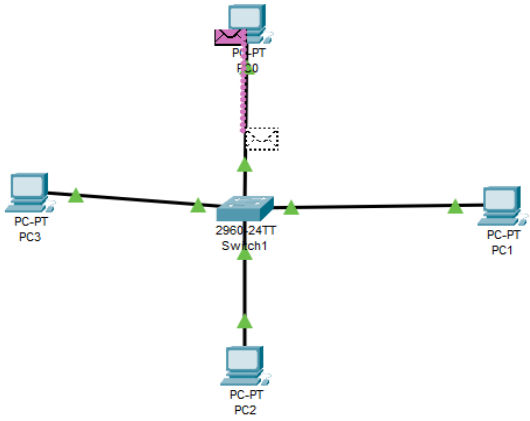


Step 5: PC1 responds the message to PC0

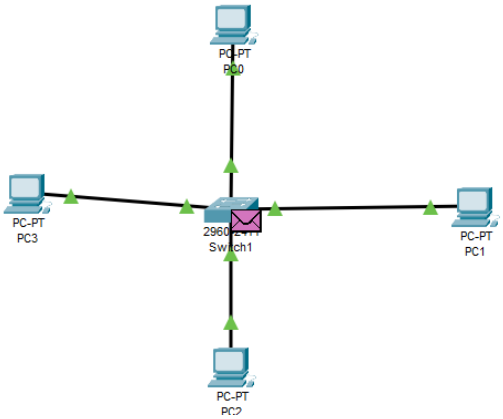


2. Switch

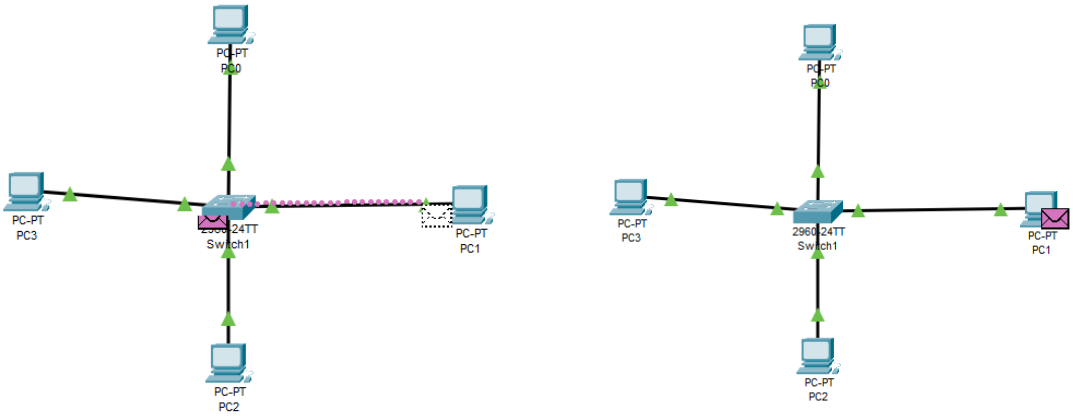
Step 1: PC0 sends a message



Step 2: Switch receives the message



Step 3: The switch checks its MAC address table and forwards the message to PC1 and then PC1 accepts the message.

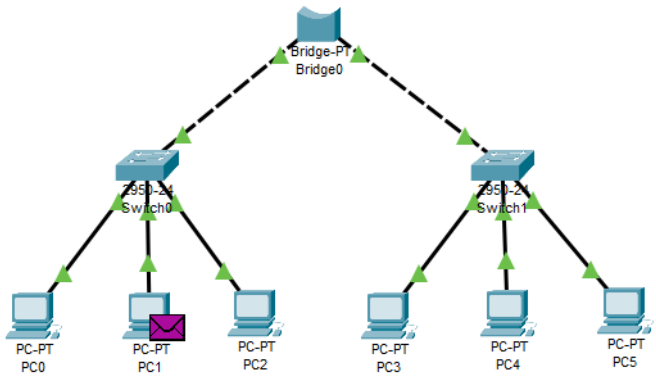


3. Bridge

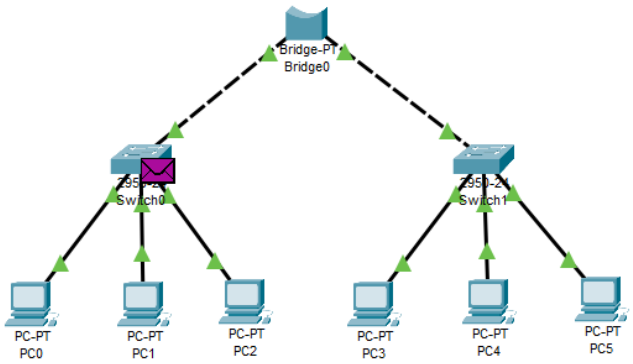
Step 1: Ping to check if connection works.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time=5ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms

C:\>
```

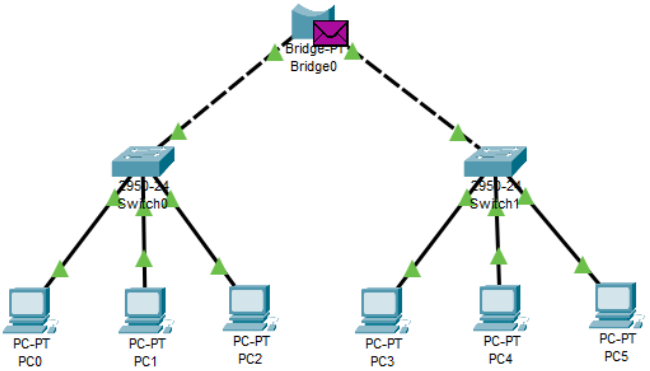
Step 2: Ready to send message.



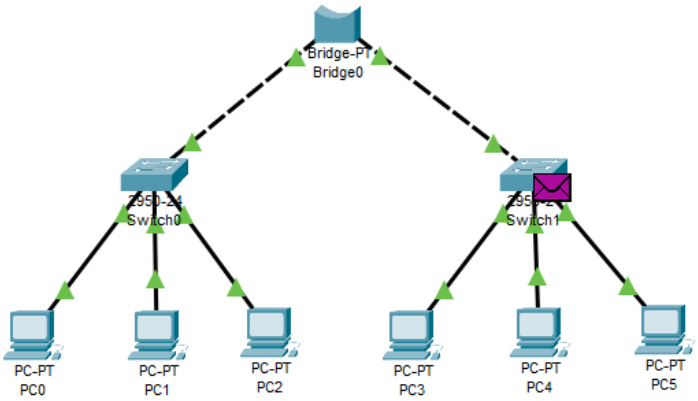
Step 3: Message sent to switch first.



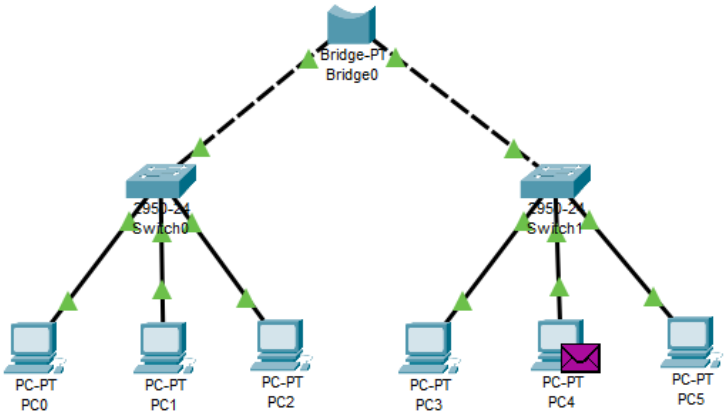
Step 4: Switch sends the message to Bridge.



Step 5: The Bridge forwards the message to the switch connected to the destination computer.



Step 6: The switch now forwards the message to the destination computer.

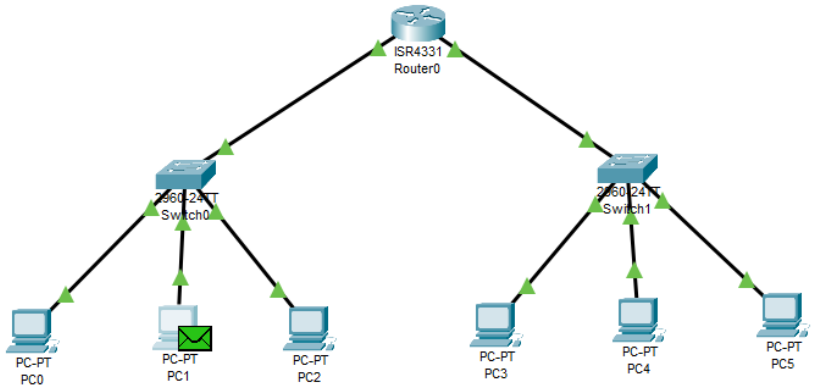


4. Router

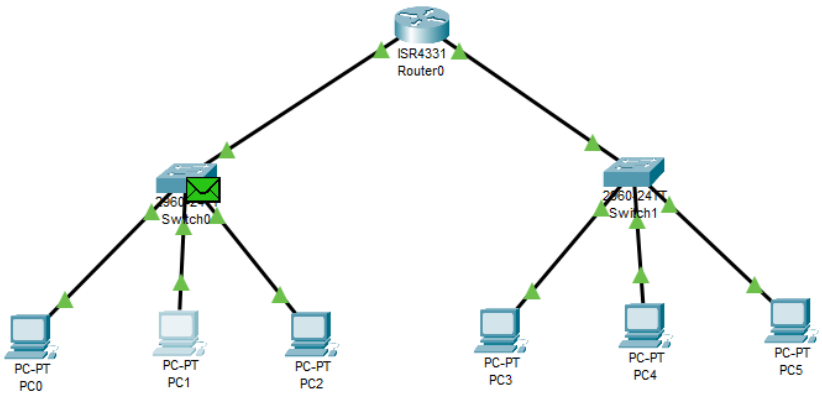
Step 1: Ping to check if connection works

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Request timed out.
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127
Reply from 10.0.0.10: bytes=32 time<1ms TTL=127

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

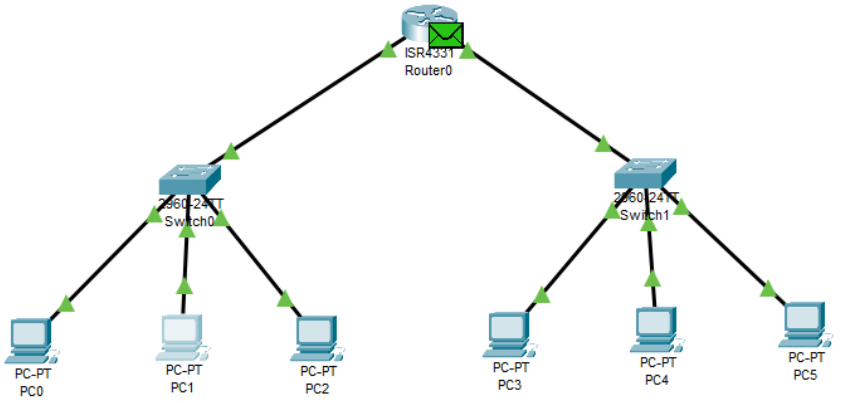
Step 2: Message ready to send



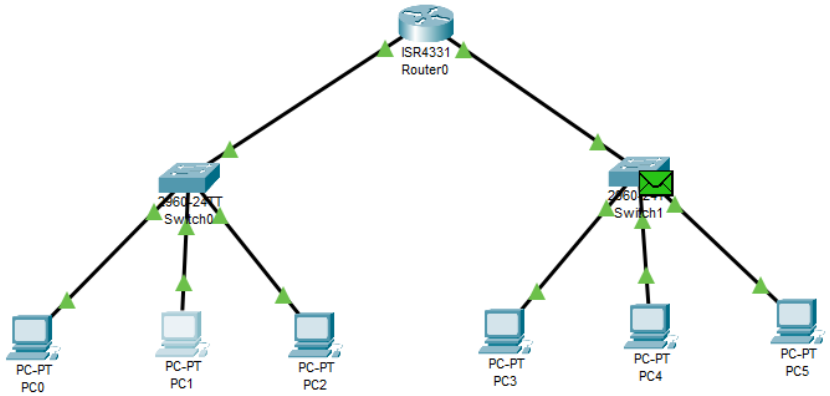
Step 3: Source sends the message to the switch it's connected to.



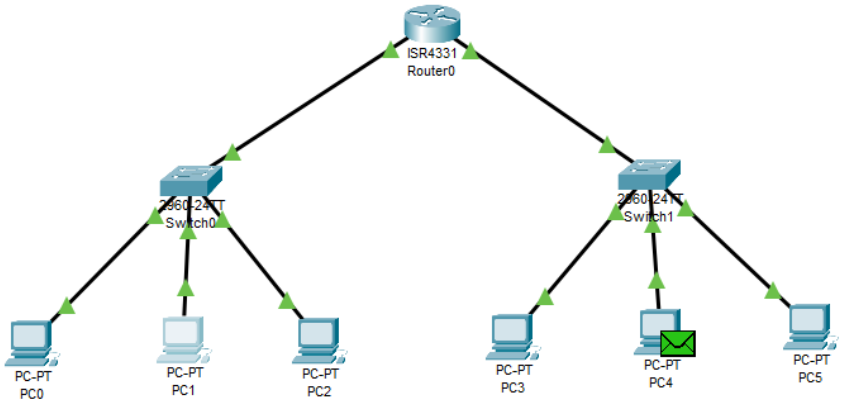
Step 4: Switch sends the message to the router.



Step 5: Router sends to the switch that the destination is connected to.



Step 6: Then the message is received my the destination computer.

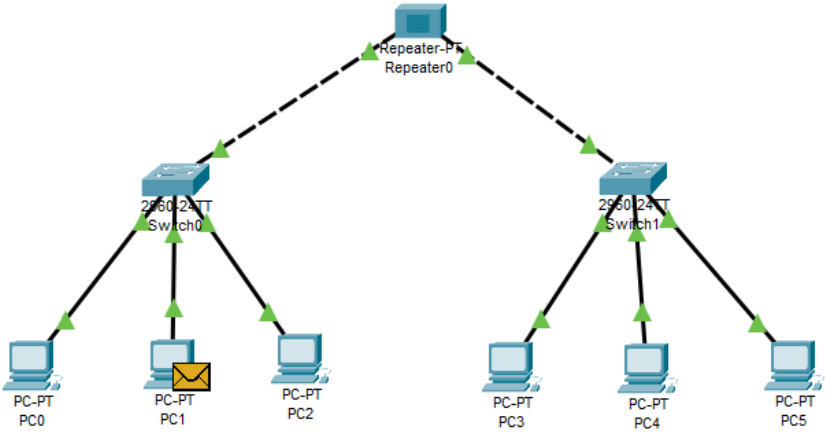


5. Repeater

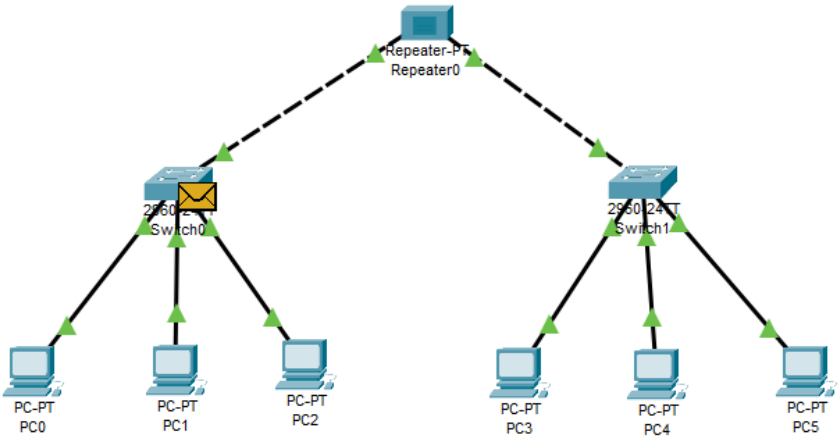
Step 1: Ping to check the connection.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.5

Pinging 192.168.1.5 with 32 bytes of data:

Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time<1ms TTL=128
Reply from 192.168.1.5: bytes=32 time=5ms TTL=128
Reply from 192.168.1.5: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 5ms, Average = 1ms
```

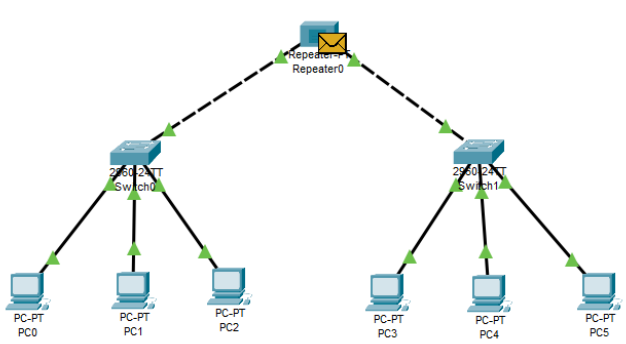
Step 2: Message ready to send.

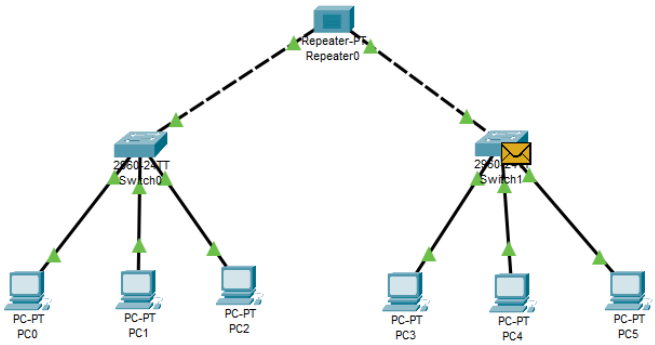


Step 3: Source sends the message to the switch it's connected to.

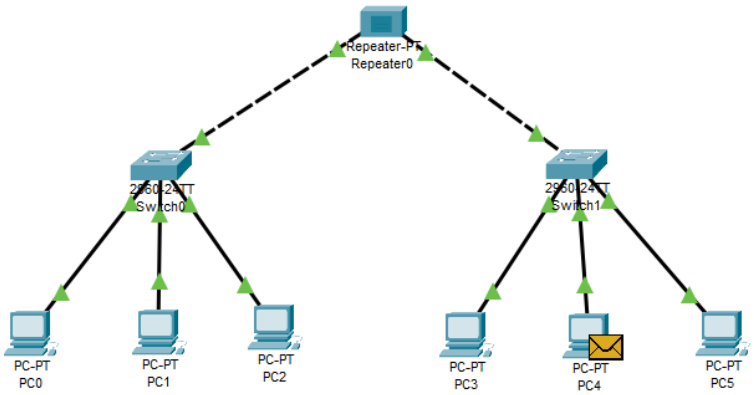


Step 4: Repeater receives the message and sends it to the switch destination computer is connected to.





Step 5: Then the message is received by the destination computer.



DISCUSSION:

In this experiment, various basic networking devices such as **hub, switch, bridge, repeater, and router** were studied and implemented using Cisco Packet Tracer. Each device was analyzed based on its function and role in data communication.

The **hub** and **repeater** were observed to operate at the **physical layer**, where they simply forward or regenerate signals without understanding the data. These devices help extend the network but do not control traffic or reduce collisions. The **switch** and **bridge**, operating at the **data link layer**, were used to forward data based on MAC addresses, which improved network efficiency by reducing unnecessary traffic.

The **router**, operating at the **network layer**, was used to connect two different IP networks. By assigning different IP addresses to each router interface and configuring default gateways on the PCs, successful communication between the networks was achieved. This

demonstrated how routers enable inter-network communication, unlike switches and hubs, which only operate within a single network.

During the experiment, practical observations such as gateway configuration, interface activation, and packet transmission behavior were noted. In some cases, limitations of the simulation environment were observed, especially while using repeaters, which helped reinforce the theoretical understanding of how devices behave differently in real networks and simulations.

CONCLUSION:

This experiment provided practical understanding of the working principles of basic networking devices including **hub, switch, bridge, repeater, and router**. It was observed that devices like hubs and repeaters extend the physical network, switches and bridges manage traffic within the same network, and routers enable communication between different IP networks using interface IP addresses and default gateways.

Overall, the experiment helped in understanding the role of each networking device, their OSI layer operation, and their importance in designing efficient computer networks.